Applied Statistics - Assessment 1

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Task 1

Task 2

a. Does Thyroid-stimulating hormone differ between male and female participants?

Thyroid-stimulating hormone *TSH* is a continuous variable while *gender* is binary. So, the independent t-test will be used and the assumptions for that test will be checked first. The following hypothesis is considered:

- H_0: There is no difference in the mean thyroid-stimulating hormone levels between male and female participants in the population
- H_0: There is a difference in the mean thyroid-stimulating hormone levels between male and female participants in the population

The assumptions to check are:

- Independent groups
- TSH levels are normally distributed in both groups
- Variance of the TSH levels are equal between both groups

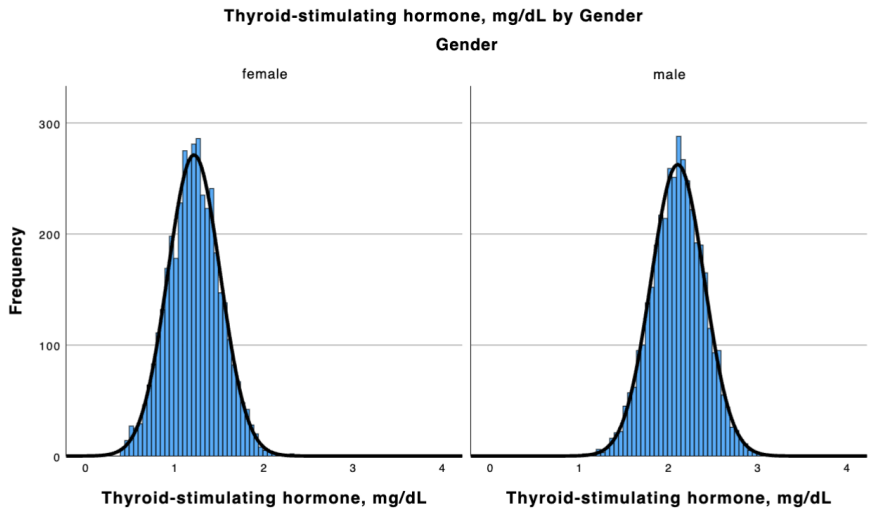


Figure 1: Distribution of TSH between male and female

All the assumptions are met, both groups are independent of one another, the TSH variable is normally distributed in both groups, and the Levene's test ($p = 0.579 > 0.05$) indicates that the variance is homogeneous.

Since the p-value is significant ($p < 0.001$), the H_0 is rejected. There is a difference in the mean thyroid-stimulating hormone levels between male and female participants in the population.

Homogeneity of Variance Test

		Levene's Test for Equality of Variances	
		F	Sig.
Thyroid-stimulating hormone, mg/dL	Equal variances assumed	.307	.579

Figure 2: Levene's test for equality of variance of TSH

Independent Samples Test

		t-test for Equality of Means							
				Significance				95% Confidence Interval of the Difference	
		t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Thyroid-stimulating hormone, mg/dL	Equal variances assumed	-133.151	7924	<.001	<.001	-.886	.007	-.899	-.873
	Equal variances not assumed	-133.135	7915.677	<.001	<.001	-.886	.007	-.899	-.873

Figure 3: Independent t-test result

In conclusion, female participants had lower thyroid-stimulating hormone levels with an average difference of 0.886 mg/dL. Furthermore, the 95% confidence interval $(-0.899, -0.873)$ indicates that mean thyroid-stimulating hormone levels in the female population is between 0.873 and 0.899 lower than that of the male population.

b. Does the level of Triglycerides differ among different levels of physical activity?

Triglycerides *triglycerides* is a continuous variable while physical activity *physical_activity* is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the distribution of triglycerides across all physical activity groups in the population
- H_1: There is a difference in the distribution of triglycerides across all physical activity groups in the population

Initially, the first test to consider is ANOVA. So the following three assumptions are checked.

- The three physical activity groups are independent
- The distribution of *triglycerides* is normal in the three groups
- The variance of *triglycerides* is equal between the three groups

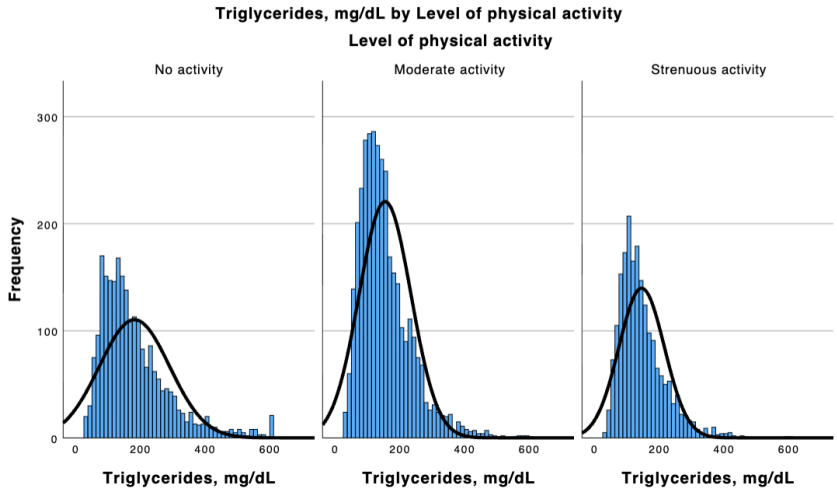


Figure 4: Distribution of triglycerides between the three physical activity groups

The first assumption is met but the figure above shows that the second assumption fails because the distribution of triglycerides for all the three groups are not normal and instead are right-skewed. Instead of the regular ANOVA, the Kruskal-Wallis test is used and the hypothesis is adjusted for non-parametric. Before the test results, the median and the IQR can be considered.

Activity Levels	Median	Lower Quartile	Upper Quartile
No Activity	153.50	104.00	228.00
Moderate Activity	137.50	97.00	192.00
Strenuous Activity	130.00	97.00	178.00

Table 1: Median and IQR of triglycerides levels across physical activity

The median from the three groups show that the triglycerides levels are generally lower as the level of physical activity gets more strenuous. But the Kruskal-Wallis test will be used to determine its significance.

Hypothesis Test Summary

	Null Hypothesis	Test	Sig.[a,b]	Decision
1	The distribution of Triglycerides, mg/dL is the same across categories of Level of physical activity.	Independent-Samples Kruskal-Wallis Test	<.001	Reject the null hypothesis.

a. The significance level is .050.

b. Asymptotic significance is displayed.

Figure 5: Kruskal-Wallis test result

The test result shows that the difference is significant ($p < 0.001$). The null hypothesis is rejected.

In conclusion, the distribution of triglycerides is different across the physical activity groups of no activity, moderate activity, and strenuous activity in the population. The triglycerides levels are lower as more intense the physical activity is.

c. Is there an association between physical activity and gender?

d. Does low-density lipoprotein differ among different BMI categories?

Low-density lipoprotein *LDL* is a continuous variable while BMI categories *BMI_cat* is a categorical variable. The hypothesis for this question are the following:

- H_0: There is no difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population
- H_1: There is a difference in the mean of low-density lipoprotein levels between the BMI categories: under & normal weight, overweight, and obese in the population

Since this is a continuous against a categorical variable, ANOVA will be used to determine the significance. But, the assumptions will be checked first for ANOVA:

- The BMI categories are independent
- The distribution of ldl is normal for the three BMI categories
- The variance of ldl is equal for the three BMI categories

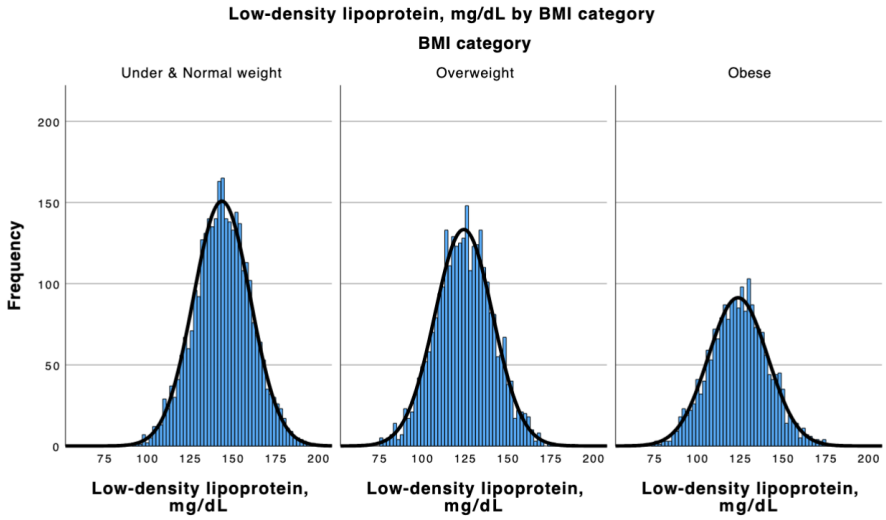


Figure 6: Distribution of ldl between the three BMI categories

The first assumption is immediately met. Figure 4 shows that the distribution of LDL across the three BMI categories are normal and Figure 5 for the result of Levene's test of homogeneity of variance ($p = 0.228 > 0.05$) indicates

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.[a]
Low-density lipoprotein, mg/dL	Based on Mean	1.479	2	7929	.228
	Based on Median	1.456	2	7929	.233
	Based on Median and with adjusted df	1.456	2	7921.631	.233
	Based on trimmed mean	1.485	2	7929	.227

a. Confidence Interval: 95%

Figure 7: Levene's test for equality of variance of ldl

that the variance is equal across the three BMI categories. All three assumptions for ANOVA is met.

ANOVA

Low-density lipoprotein, mg/dL

	Sum of Squares	df	Mean Square	F	Sig.[a]
Between Groups	722980.139	2	361490.069	1271.414	<.001
Within Groups	2254382.852	7929	284.321		
Total	2977362.991	7931			

a. Confidence Interval: 95%

Figure 8: ANOVA results

Multiple Comparisons

Dependent Variable: Low-density lipoprotein, mg/dL
Scheffe

(I) BMI category	(J) BMI category	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Under & Normal weight	Overweight	19.322*	.437	<.001	18.25	20.39
	Obese	19.810*	.485	<.001	18.62	21.00
Overweight	Under & Normal weight	-19.322*	.437	<.001	-20.39	-18.25
	Obese	.488	.495	.615	-.72	1.70
Obese	Under & Normal weight	-19.810*	.485	<.001	-21.00	-18.62
	Overweight	-.488	.495	.615	-1.70	.72

*. The mean difference is significant at the 0.05 level.

Figure 9: Post-hoc Scheffe test

In conclusion, The ANOVA test ($p < 0.001$) indicates that the H_0 is rejected. Thus, there is a difference in the mean of low-density lipoprotein across the three BMI categories under & normal weight, overweight, and obese in the population. Furthermore, the post-hoc test shows the significant difference in the mean between the groups:

- Under & normal weight and overweight is around 19.3
- Under & normal weight and obese is around 19.8

While the difference between the overweight and obese groups is not statistically significant.

e. Is there an association between smoking and CAD?

f. Is there an association between congenital anomalies and CAD?

g. Is there a change in Total cholesterol between baseline and after intervention?

Total cholesterol base and after (*Tchol_base* and *Tchol_after*) are continuous variables and since it wants to know if the change between baseline and after, the paired t-test will be considered and the assumption *The difference between total cholesterol at baseline and after intervention is normally distributed* will be checked first and the following hypothesis will be used:

- H_0: The mean total cholesterol between baseline and after intervention is the same in the population
- H_1: The mean total cholesterol between baseline and after intervention is not the same in the population

The new variable *diff_tchol* is introduced by subtracting *Tchol_base* from *Tchol_after*

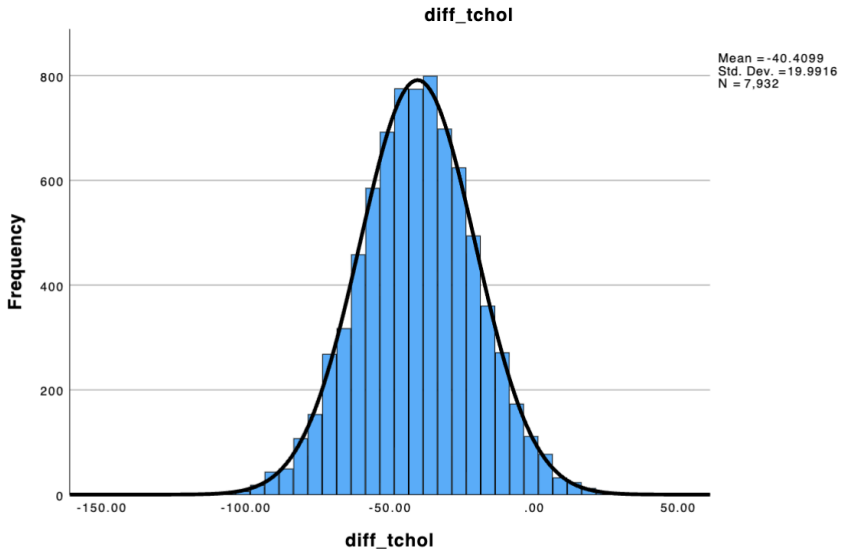


Figure 10: Difference in total cholesterol at baseline and after intervention

The figure above shows the new variable is normally distributed. So, the assumption is met.

Paired Samples Test

		Paired Differences					t	df	Significance	
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			One-Sided p	Two-Sided p
Pair 1	Total cholesterol at baseline, mg/dL - Total cholesterol after follow-up, mg/dL	40.410	19.992	.224	39.970	40.850	180.024	7931	<.001	<.001

Figure 11: Paired t-test

The t-test result ($p < 0.001$) shows that it is significant and that the H_0 should be rejected.

In conclusion, the mean total cholesterol between baseline and after intervention is different in the population. The average total cholesterol after intervention is around 40 mg/dL lower. Whereas the 95% confidence interval for the mean difference is $(39.97, 40.85)$ indicating that the mean total cholesterol after intervention in the population is between 39.97 mg/dL lower or 40.85 mg/dL higher.